

COMP 6600 - Mid-Project Report

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Abstract

Accurate satellite orbit prediction underpins navigation, communication, and Earth observation. Traditional methods like the Simplified General Perturbation 4 (SGP4) model are efficient but accumulate error over time due to simplified force modeling and unmodeled perturbations. With the rise of precise tracking data, data-driven techniques now offer a way to enhance prediction accuracy.

This project develops a two-phase hybrid framework combining physics-based propagation and machine learning correction for the LAGEOS-1 satellite. Using Two-Line Element (TLE) data from CelesTrak and precise SP3 ephemerides from NASA Earthdata.

1. Introduction

The problem of accurately estimating satellite positions is critical in modern navigation, communication, and Earth observation systems. Satellite position information underpins applications such as Global Navigation Satellite Systems (GNSS), weather prediction, and satellite-based internet services. Traditionally, position estimation relies on physics-based orbital mechanics models or Kalman filtering techniques, which require detailed physical parameters and can be sensitive to modeling errors. With the growing availability of large-scale satellite tracking data, neural networks provide a promising alternative, capable of learning complex nonlinear relationships between orbital parameters and satellite positions.

1.1. Problem Statement

The core problem is to build a model that can predict satellite positions at future time steps, given past trajectory data or orbital parameters. Specifically, the dataset will consist of publicly available satellite ephemeris data. The expected result is a neural network model that can accurately estimate the position of satellites in Earth-Centered Earth-Fixed (ECEF) coordinates.

The method proposed is to develop and train a neural network architecture that can capture temporal dependencies in

satellite trajectories. While existing implementations for orbital propagation (such as SGP4 models) are widely used, the plan is to use them both as a source of labeled training data and as baselines for comparison. Improvement over these baseline models will be sought by leveraging the neural network's ability to learn nonlinear error patterns and generalize to different orbital regimes.

1.2. Problem Approach

The overall approach for this project is a two-phase machine learning framework designed to improve orbit prediction accuracy by learning the residual errors between physics-based and observed satellite trajectories. The project will utilize LAGEOS-1 satellite data, combining Two-Line Element (TLE) sets with precise orbit ephemerides. The goal is to train a machine learning model capable of forecasting and correcting the systematic propagation errors of the SGP4 model, thus improving the long-term prediction accuracy.

In the first phase, TLE and SP3 datasets will be processed to extract satellite positions and velocities over matching time intervals. The SGP4 propagator will generate predicted orbits from the TLEs, while the SP3 data will serve as ground truth. The difference between these datasets will form the training data for the machine learning model.

The second phase involves training and applying a time-delay neural network (TDNN) to model the error evolution over time. The network will use a convolutional layer to extract temporal features, followed by dense layers to forecast future error corrections. Once trained, the TDNN will predict SGP4 error trends and apply these corrections recursively to improve orbit forecasts.

2. Background and Related Work

2.1. Satellites Orbit Information

Orbital information for Earth satellites is typically represented using the Two-Line Element (TLE) format, which encodes mean orbital elements derived from radar and optical tracking by the U.S. Space Surveillance Network (SSN).

TLEs are publicly distributed by the U.S. Strategic Command and are widely used due to their accessibility and standardized structure. Each TLE represents a mean Keplerian state corrected for perturbative effects such as Earth's oblateness and atmospheric drag.

However, while TLEs provide a convenient global catalog, they are not inherently precise. Their accuracy depends on the underlying fitting model, observation density, and update frequency. Studies such as Wei and Zhao [2] and Aida and Kirschner [1] have demonstrated that TLE-based position estimates can deviate by hundreds of meters to several kilometers, depending on altitude and propagation time. These inaccuracies make raw TLE data insufficient for applications requiring high-precision orbital prediction, such as collision avoidance and autonomous navigation.

2.2. Orbit Propagators

Orbit propagation models transform an initial orbital state forward in time to estimate future positions and velocities. These propagators generally fall into two categories: numerical and analytical.

Numerical propagators, such as the High-Precision Orbit Propagator (HPOP), integrate the equations of motion using detailed physical models that include high-order gravity terms, solar radiation pressure, atmospheric drag, and third-body perturbations. They achieve centimeter- to meter-level accuracy when properly initialized but are computationally intensive and require detailed spacecraft parameters that are often unavailable for public datasets.

Analytical propagators, such as the Simplified General Perturbation models (SGP4/SDP4), provide a computationally efficient alternative. They approximate orbital motion using mean element models and are the standard for propagating TLEs. However, their simplified physics introduces cumulative errors over time.

Extensive evaluations have quantified SGP4's accuracy across orbital regimes. Wei and Zhao [2] analyzed 1,120 space objects and reported orbit determination errors on the order of 100 m for low-Earth orbits (LEO), 0.7 km for medium-Earth orbits (MEO), and 1.9 km for geosynchronous (GEO) satellites, with typical prediction errors remaining under 40 km for up to 30-day propagations. Aida and Kirschner [1] compared SGP4 outputs with GPS-based reference orbits from GRACE-1, TerraSAR-X, and TET satellites, finding that along-track errors dominate and grow approximately quadratically with time, reaching tens of kilometers after a week. These results underscore SGP4's efficiency but also its inherent limitations in precision and long-term reliability.

2.3. Use of Deep Learning in Orbit Propagation

With the growing availability of precise ephemerides from missions and commercial constellations, data-driven methods have emerged as powerful tools for orbit prediction. Traditional physics-based models can struggle with unmodeled perturbations, atmospheric variability, and non-conservative forces, whereas machine learning (ML) techniques can approximate complex nonlinear relationships directly from observed data. These approaches are especially relevant in low-Earth orbit (LEO), where drag, solar activity, and density fluctuations introduce highly nonlinear effects that are difficult to represent with simplified analytical propagators such as SGP4.

Research in this field has primarily followed two main strategies: direct orbital prediction and residual error correction. Although both rely on learning temporal dynamics from data, they differ in scope, data representation, and their reliance on physical models.

In the direct orbital prediction approach, a neural network is trained to forecast future satellite states typically position and velocity vectors directly from past orbital data. This method treats orbit propagation as a sequence forecasting problem, relying entirely on data to learn the underlying motion. The key advantage is flexibility: the network can, in principle, learn complex environmental dependencies without requiring explicit modeling of physical forces. However, this also introduces drawbacks. Because no physical constraints are imposed, prediction errors can accumulate rapidly, leading to drift or instability during long-term propagation. Models trained in this manner also tend to generalize poorly when exposed to conditions or orbital regimes not represented in the training set.

The second strategy, known as residual error correction, combines the strengths of analytical propagation and machine learning. In this hybrid framework, a baseline physics-based propagator such as SGP4 provides the nominal orbit prediction, while an ML model is trained to learn the residual discrepancy between this analytical output and high-accuracy reference data (for instance, from numerical propagators or precise ephemerides). This approach effectively offloads the burden of modeling well-understood dynamics to the analytical propagator, allowing the ML model to focus on compensating for unmodeled perturbations such as drag variability, solar flux changes, and higher-order gravitational effects. Because the network operates on smaller residual magnitudes rather than absolute position values, it generally requires less data and converges more efficiently. Importantly, this formulation preserves the physical interpretability of the baseline model while improving its overall accuracy.

A representative example of this hybrid approach is presented by Kassis et al.[4], who trained neural networks to map SGP4 outputs to high-fidelity trajectories generated by

the High-Precision Orbit Propagator (HPOP) for Orbcomm satellites. By learning the residual error between SGP4 and HPOP, their *SGP4 + ML* framework achieved sub-10-meter position accuracy while maintaining real-time computational efficiency.

2.4. Time-Dependent Neural Networks

A wide range of neural network architectures have been investigated for orbit prediction and estimation, each with varying levels of success depending on the data structure and temporal complexity of the problem. The IPA Machine Learning for Orbit Determination study [3] and the Kassas study [4] both compared architectures such as feedforward neural networks (FNNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs) to assess their suitability for learning satellite motion dynamics. These studies consistently found that models capable of representing temporal dependencies outperform static networks that process each orbital state independently. This is because orbital motion, especially in low-Earth orbit (LEO), exhibits strong time-correlated behavior due to periodic gravitational, atmospheric, and solar effects.

Among the time-dependent architectures tested, the Time-Delay Neural Network (TDNN) and the Long Short-Term Memory (LSTM) network have emerged as particularly effective for orbit prediction tasks. Both architectures are designed to capture sequential dependencies in time-series data, but they differ in how they represent and retain temporal information. The TDNN extends a traditional feedforward neural network by incorporating delayed versions of the input signal, allowing the network to learn temporal correlations over a fixed window of previous states. This makes the TDNN especially well-suited for short-term orbital forecasting, residual correction, and real-time applications where the model must operate efficiently with limited computational resources. Kassas demonstrated that TDNN-based hybrid models, when used to learn residual errors between SGP4 and high-fidelity HPOP trajectories, significantly reduced propagation error while maintaining real-time performance [4].

The LSTM, by contrast, is a recurrent architecture that employs gated memory cells to selectively retain or forget information over longer sequences. This capability enables LSTMs to capture complex, long-term dependencies in orbital behavior, such as gradual drift due to atmospheric drag or solar activity cycles. The IPA study showed that LSTMs generalize better than feedforward or convolutional models when trained on noisy or sparse orbital datasets, as their recursive design naturally models temporal continuity and error accumulation. LSTMs are therefore particularly beneficial for predicting long-duration orbits and learning slow-varying perturbations that cannot be captured effectively by shallow or static models.

3. Project Overview

The overall approach for this project is a two-phase machine learning framework designed to improve orbit prediction accuracy by learning the residual errors between physics-based and observed satellite trajectories. Specifically, the project will utilize LAGEOS-1 satellite data, combining Two-Line Element (TLE) sets from CelesTrak with precise orbit ephemerides in SP3 format obtained from NASA's Crustal Dynamics Data Information System (CD-DIS), a database that archives space geodesy data. The goal is to train a machine learning model capable of forecasting and correcting the systematic propagation errors of the SGP4 model, thus improving the long-term prediction accuracy.

The first phase of the framework focuses on the data collection and preparation. The two datasets, TLE and SP3, will be processed to extract the corresponding satellite positions and velocities over identical time intervals. The SGP4 propagator will be used to compute positions from the TLEs, while the SP3 data will serve as the truth reference. The difference between these two data sources will form the primary input to the learning model. This "error-based" training strategy is critical because it allows the model to learn the discrepancies between the well-understood propagators and truth reference, rather than directly modeling the orbital dynamics themselves. This is beneficial because the orbital mechanics are governed by complex and nonlinear perturbative forces. This approach effectively isolates the behavioral differences between the analytical and observed trajectories, reducing the degree of nonlinearity the network must learn.

The second phase involves developing and training the machine learning model. The TDNN will include an initial convolutional layer to extract temporal dependencies between consecutive orbital samples, followed by pooling and fully connected layers that output predicted error corrections. Hyperparameters will be chosen based on relevant literature.

Once trained, the TDNN will be used to forecast SGP4 propagation errors and apply corrections to future orbit predictions. During this operational phase, the latest TLE and SP3 data will be retrieved, and the network will use the most recent error samples to predict future correction terms. The corrected position and velocity estimates will then be obtained by adding the model's predicted errors to the SGP4-propagated outputs.

The performance of the framework will be evaluated through trajectory comparison plots and statistical error metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and orbit-specific components including radial, in-track, and cross-track errors. Improvements over baseline SGP4 propagation will demonstrate the capability of the neural model to generalize across time and

provide enhanced prediction fidelity.

4. Preliminary Results

For the preliminary stage of this work, TLE data for the LAGEOS-1 satellite were obtained from CelesTrak and propagated using the SGP4 model. Corresponding SP3 precise ephemerides from NASA's Earthdata database were used as reference truth data. The two datasets were time-aligned, and position errors were computed in the Earth-Centered Inertial (ECI) frame. The following figures show the comparison between the propagated orbit and the truth data, along with the resulting position errors that will be used as inputs for the neural network.

Figure 1 shows the comparison between the SGP4-propagated orbit and the SP3 truth trajectory. The two orbits overlap closely, demonstrating that SGP4 provides a good approximation of the satellite's motion. Small deviations become visible over time as propagation errors accumulate.

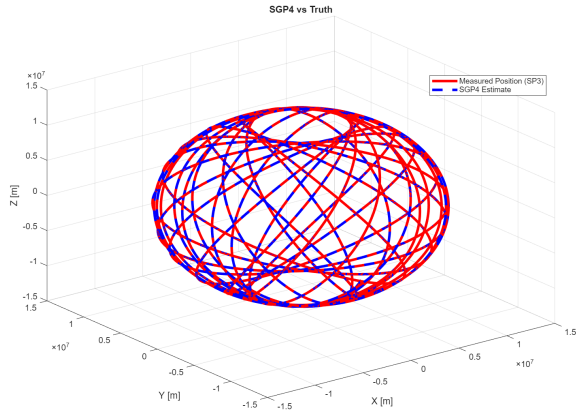


Figure 1. Comparison of the SGP4-propagated orbit (blue) and SP3 truth orbit (red) in the ECI frame.

Figure 2 presents the component-wise position errors in the X, Y, and Z directions within the ECI frame. The periodic oscillations in each component correspond to the orbital period, showing how the error varies predictably over time. These fluctuations are largely driven by unmodeled perturbations such as atmospheric drag variability and solar radiation effects.

Figure 3 shows the magnitude (norm) of the total position error over time. The overall error typically remains within a few hundred meters, with peaks occurring periodically as the orbit evolves. This trend is consistent with expected SGP4 accuracy over multi-day propagation and provides a baseline error level for the neural network to learn.

References

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Figure 2. Component-wise position errors (X, Y, Z) between SGP4 and SP3 orbits in the ECI frame.

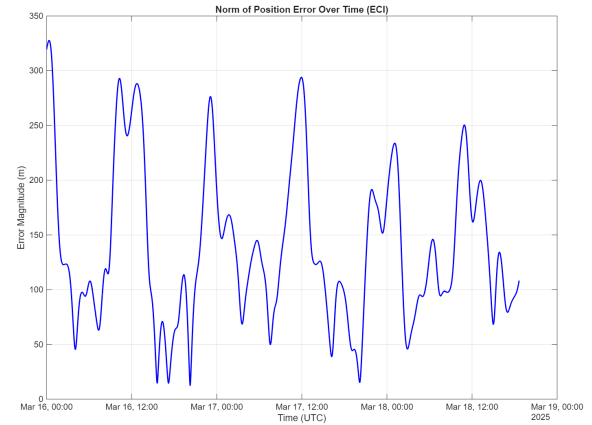


Figure 3. Norm of the total position error between SGP4 and SP3 over time in the ECI frame.

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